# Pragnidhved Reddy Pinnamareddy

ee23btech11050@iith.ac.in | pragnidhved@gmail.com | +91 9398140514 | LinkedIn | GitHub

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.16 |
| Sri Chaitanya Junior Kalasala | Class XII (TSBIE) | 2023 | 97.50% |
| Kennedy High The Global School | Class X (CBSE) | 2021 | 93.20% |

## WORK EXPERIENCE

**Deccan AI -- AI Data Training & Evaluation Freelancer**

- Contributed to **4 AI training projects since March 2026**, assessing LLM-generated responses across coding, reasoning, SQL, and human-interpretability tasks to deliver detailed post-training feedback.
- Evaluated **LLMs and AI agents** on reasoning quality, code correctness, tool usage, instruction adherence, and task completion, identifying shortcomings and applying task-specific evaluation criteria.
- Improved **human-generated training data** through annotation, response refinement, error identification, and multi-stage quality validation to support iterative model training and evaluation.

## PROJECTS

**Document Q&A Chatbot using RAG** | Python, FAISS, Sentence Transformers, Gemini

- Built a **Retrieval-Augmented Generation (RAG)** chatbot supporting PDF, TXT, and CSV documents with automated text extraction, preprocessing, and overlapping chunk generation.
- Generated semantic embeddings using **Sentence Transformers (all-MiniLM-L6-v2)** and implemented **FAISS L2 similarity search** to retrieve the top-4 relevant document chunks for each user query.
- Integrated **Gemini 2.5 Flash** to generate context-grounded responses from retrieved document content, with source attribution through an interactive **Gradio** chat interface.

**Pet Breed Classification & Counterfactual Image Generation** | PyTorch, ResNet-18, Computer Vision

- Fine-tuned a pretrained **ResNet-18** model for 37-class pet breed classification on the **Oxford-IIIT Pet dataset**, using image preprocessing, supervised training, and GPU acceleration.
- Developed an **optimization-based counterfactual generation pipeline** that modifies input images to shift model predictions toward randomly selected target classes while minimizing unnecessary visual changes.
- Combined **target-class cross-entropy, L1, total-variation, and blur losses** to constrain perturbations, and visualized counterfactuals using **difference maps, threshold maps, and overlay analysis**.

**Communication-Efficient Distributed Learning for Convex Logistic Regression** | Distributed Optimization, Logistic Regression

- Developed a **distributed logistic regression framework** across 100 edge devices, aggregating local model weights at a central base station under a fixed communication budget for efficient distributed training.
- Modeled **Rayleigh fading, path loss, TDMA transmission, and retransmission failures**, deriving a KKT-based allocation strategy to maximize the overall probability of successful device participation.
- Solved the allocation constraint using **bisection**, converted continuous solutions into integer trial budgets, and evaluated model accuracy across communication rounds on **MNIST under unreliable communication conditions**.

**Adaptive Noise Cancellation** | MATLAB, RLS, Digital Signal Processing

- Engineered an **adaptive noise-cancellation pipeline using Recursive Least Squares (RLS)** filtering to estimate noise from a reference signal and generate a cleaner speech output.
- Implemented **online coefficient adaptation, inverse correlation-matrix updates, and configurable frequency-selective filtering** for effective suppression of dynamically varying noise components.
- Developed **automated SNR-based evaluation and signal visualization** to quantitatively compare noisy, reference, and denoised speech signals across different filtering conditions.

## SKILLS

- **Programming Languages :** C, C++, Python, SQL, MATLAB, Verilog
- **Tools :** LT Spice, nanoHUB, Solid Edge, Fusion360
- **Others :** Data Structures and Algorithms, Latex, Git, Github, Jupyter Notebook
- **Soft Skills :** Decision Making, Time Management, Adaptability, Communication

## RELEVANT COURSES

**Computer Science Courses :** Introduction to Programming, Introduction to AI, Introduction to Modern AI, Pattern Recognition and Machine Learning, Data Structures and Algorithms, Convex Optimization.

**Math Courses :** Calculus, Matrix Theory, Probability and Random Processes, Complex Variables, Vector Calculus, Differential Equations, Predictive Modelling.

**Core Courses :** Digital Systems, Signals and Systems, Digital Signal Processing, Control Systems, Analog Electronics, Semiconductor Device Fundamentals, Power System Analysis, Electrical Machines, Statistical Inference, Information Theory, Digital IC Design

## POSITIONS OF RESPONSIBILITY

**Events & Hospitality Coordinator -- MILAN, IIT Hyderabad**

- Contributed to the planning, coordination, and successful execution of MILAN, supporting event operations and hospitality activities.
- Coordinated with teams and stakeholders to ensure smooth execution of event-related activities and on-ground operations.
- Assisted in managing event logistics and hospitality arrangements, contributing to an organized and seamless experience for participants and guests.

## EXTRACURRICULAR

- Competed as a **Setter** in the **U-19 State-Level Volleyball Clusters**, representing the team at the state level, helping secure **3rd place**, and earning the **Best Setter of the Tournament** award for outstanding performance.
- **Represented Telangana at the National-Level Sepak Takraw Championship on three occasions**, competing against teams from across the country and gaining repeated experience at the national level in a competitive sporting environment.
- **Participated in MILAN Inter-Hostel Gaming Competitions**, representing my hostel in **PC and mobile gaming events**, and secured **2nd place in the Call of Duty competition**.